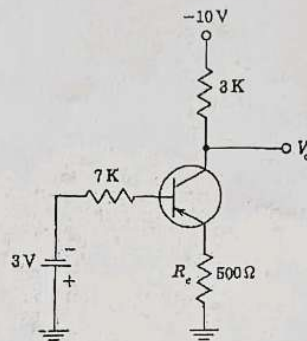
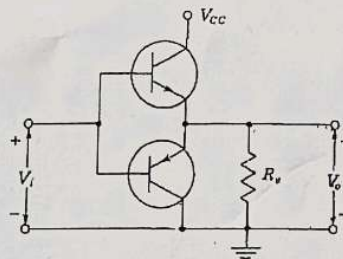


1. Discuss energy band theory of crystals and its implications with the help of neat illustrations.
2. For the circuit shown in the Fig. below, assume $\beta = h_{FE} = 100$. Find i) if the silicon transistor is in cutoff, saturation or in the active region ii) V_o iii) the minimum value for the emitter resistor R_e for which the transistor operates in the active region



3. The circuit shown below is an amplifier using a p-n-p and an n-p-n transistor in parallel. The two transistors have identical characteristics. Find the expression for the voltage gain and the input resistance of the amplifier, using the simplified hybrid model.



4. Prove that $Y_o = h_o \left(\frac{R_s + R_{i0}}{R_s + R_{i0}} \right)$ where $R_{i0} \equiv R_i$ for $R_L = \infty$ and $R_{i0} \equiv R_i$ for $R_L = 0$ in the circuit shown in the Fig. below

